

SAME-ITEM PAIR RANKING FOR CROSS-CORPUS SPEECH DEEPPAKE DETECTION

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ABSTRACT

Pairwise objectives are used in speech anti-spoofing, but it is unclear whether matching a documented natural-synthetic source item, rather than an arbitrary opposite-class item, improves transfer. We test this question with a controlled three-way comparison. A compact, fresh-initialized Res2TCNGuard detector is trained on a shared ODSS source pool using (i) binary cross-entropy (B1), (ii) the same loss plus random opposite-class ranking (B2), or (iii) same-item natural-synthetic ranking (P). Source-only selection fixes all hyperparameters, checkpoints, and the four-seed ensemble before two fixed external evaluations. P obtains 47.07%/42.98%/45.02% equal error rate (EER) on SONAR/ArAD/macro, versus 58.88%/44.69%/51.78% for B1 and 56.09%/47.93%/52.01% for B2. Paired trial-bootstrap intervals, conditional on the frozen ensemble, are P-B1 $[-8.25, -5.25]$ and P-B2 $[-8.54, -5.30]$ percentage points. Thus, under this protocol, same-item ranking lowers the predeclared two-target macro EER beyond an equal-edge random-pair control. The documented key is not a transcript-level semantic annotation; we make no state-of-the-art or causal representation claim.

Index Terms— speech deepfake detection, anti-spoofing, pairwise ranking, cross-corpus generalization, controlled evaluation

1. INTRODUCTION

Speech anti-spoofing can fail when a detector is moved to a new corpus or generator family. Such shifts are often differences in data distribution, rather than simply a harder instance of the original task [1]. Pairwise learning is one route to improve a representation: a Siamese anti-spoofing system learns class agreement between two samples [2], supervised contrastive learning has been combined with cross-entropy for spoof detection [3], and bonafide-pair learning aligns two augmentations of a bonafide utterance [4]. We do not claim pairwise supervision as new. Instead, we ask whether a documented same-item pairing changes transfer relative

to a fixed, equal-edge random-pair control.

We test this with naturally paired resynthesis data. ODSS provides documented natural and synthetic variants of source utterances [5]. We compare same-item ranking to two controls that preserve the source pool, network, binary objective, number of ranking edges, rank weight, optimizer, and selection procedure. The random-pair control is important: without it, a gain could be attributed to any added margin term rather than the documented same-item relation.

Our contributions are:

- a same-item paired-ranking objective and an equal-edge, stratum-matched random-ranking control for separating the documented pairing key from generic rank regularization;
- a source-only training/selection protocol with fresh initialization, voice-disjoint source development, target-label firewall, and source-target waveform-fingerprint audit; and
- a two-target, four-seed terminal evaluation in which same-item ranking lowers EER relative to both controls under a precommitted, ensemble-conditional trial-bootstrap gate.

2. CONTROLLED SAME-ITEM RANKING

2.1. Matched source pool

We use ODSS revision 1968e6d0 as the only training source. Its metadata yields 7,961 complete groups, each containing one natural recording, a VITS resynthesis, and a FastPitch/HiFiGAN resynthesis. Groups are keyed by corpus, speaker, and stem. We retain 23,883 trials (7,961 bona fide and 15,922 spoof) and remove 3,071 unmatched VITS rows from all methods. Thus BCE does not receive additional unpaired data. A deterministic language-stratified split assigns each source voice key to either train or source development.

Let $z(x)$ be a spoof-evidence logit and (x_s, x_b) a spoof-bona-fide edge. Every method receives the same class-balanced BCE batches. B1 minimizes $\mathcal{L}_{\text{BCE}}$. For

$m \in \{\text{B2}, \text{P}\}$, the objective is

$$\mathcal{L}_m = \mathcal{L}_{\text{BCE}} + \lambda \mathcal{L}_{\text{rank}}, \quad (1)$$

where

$$\mathcal{L}_{\text{rank}} = \frac{1}{|E|} \sum_{(x_s, x_b) \in E} \max\{0, 1 - [z(x_s) - z(x_b)]\}. \quad (2)$$

P uses the 15,922 natural-synthetic edges from the same documented group. B2 uses exactly the same spoof endpoints and edge count, but pairs each with a different-group bona-fide trial selected deterministically within its split/language/source-corpus/generator stratum. Thus the intended B2-P difference is same-item versus different-item pairing. The key is metadata, not a transcript or semantic-equivalence proof; it can bundle speaker, recording, duration, and phonetic correspondence with the documented item.

2.2. Frozen optimization and terminal firewall

All models are freshly initialized instances of the public 172k-parameter Res2TCNGuard architecture [6]; its released ASVspoof-pretrained checkpoint is prohibited. We use BF16 autocast, batch 24, AdamW (10^{-4} learning rate, 10^{-2} weight decay), no augmentation, and at most six epochs. A source-only grid over rank weight $\{0.1, 0.3, 1.0\}$ selects 1.0 by mean voice-disjoint development EER; that P-selected value is imposed unchanged on B2. Four fixed seeds use the lowest-development-EER checkpoint (lower epoch on a tie). No target audio, label, metric, or score selects a source choice.

The terminal targets are fixed SONAR (3,948 trials) and ArAD (3,570 trials) releases from the Speech DF Arena ecosystem [7]. After all 12 source checkpoints are hash-sealed, we materialize label-free target records and separately hash labels. The evaluator verifies every audio byte hash and canonical 16-kHz waveform fingerprint, writes all raw predictions, then opens labels once. It rejects any exact source-target fingerprint collision; none is observed. We report EER (the threshold at which false-accept and false-reject rates are equal; lower is better) and AUROC with spoof as the positive class from the mean spoof probability of the four frozen seeds. Target metrics never enter model selection; we nevertheless do not call this evaluation blind because historical target rows for the excluded public checkpoint were visible during the architecture-provenance audit.

3. RESULTS

3.1. Source-only selection

The rank weight is the only loss hyperparameter selected from source development. Before any target materializa-

Table 1. Complete source-only P rank-weight selection. Mean development EER is over four fixed seeds; lower is better. The selected value is reused unchanged for B2 and P terminal fits.

λ	0.1	0.3	1.0
Mean dev. EER (%)	33.60	25.92	22.05

Table 2. Terminal score diagnostics. Scores are mean spoof probabilities; AUROC treats spoof as positive. Target counts are bona-fide/spoof.

Target (bona/spoof)	Method	EER (%)	AUROC
SONAR (2274/1674)	B1	58.88	.411
	B2	56.09	.428
	P	47.07	.557
ArAD (484/3086)	B1	44.69	.586
	B2	47.93	.519
	P	42.98	.615

tion, we run the fixed P grid with four seeds per value and choose lower mean development EER (lower weight on a tie). Table 1 shows the complete selection evidence. The chosen $\lambda = 1.0$ is then imposed on B2 as well as P, so a terminal difference cannot be caused by a different ranking scale. B2 is deliberately not tuned at its own optimum: it is the fixed P-lambda, equal-edge control. All methods share four fresh initializations, BF16 batch 24, four workers, and at most six epochs; B1 does not execute rank-edge forwards, so this is a matched training envelope rather than equal wall-clock compute across all three methods.

3.2. Terminal external evaluation

Figure 1 reports the complete predeclared primary EER comparison. P has lower ensemble EER than B1 and B2 on both targets. Its 45.02% macro EER is 13.05% relatively lower than B1 (51.78%) and 13.44% relatively lower than B2 (52.01%). The predefined 2,000-replicate, label-stratified shared-ID trial bootstrap (seed 2909), conditional on the frozen four-seed ensemble, gives P-B1 -6.70 pp (95% CI $[-8.25, -5.25]$) and P-B2 -6.97 pp (95% CI $[-8.54, -5.30]$). Both intervals are below zero, and all precommitted terminal checks pass.

Table 2 gives score diagnostics and class counts. P has the highest AUROC on both targets, but all values and the weak absolute EERs must be read as a narrow transfer comparison rather than a deployment result.

The result should not be confused with a strong absolute detector. P remains at 47.07% EER on SONAR and 42.98% on ArAD. Further, the decision concerns a predeclared four-seed ensemble: individual-seed macro EER varies from 41.10% to 53.30% for P. The trial-bootstrap

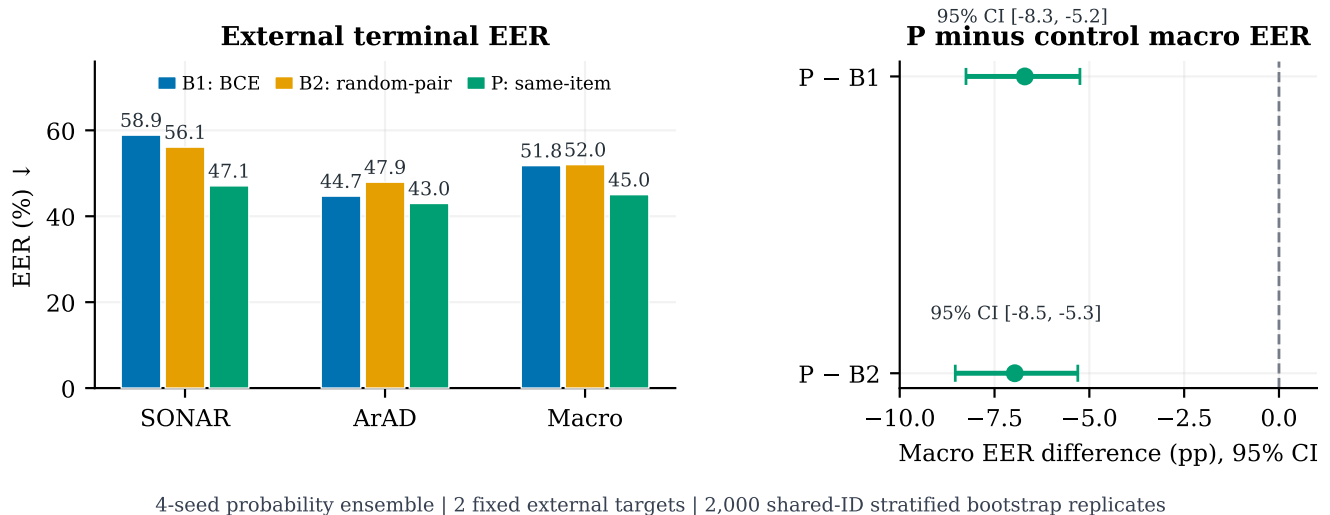


Fig. 1. Sealed terminal evaluation. Left: EER of the four-seed probability ensemble on the two fixed external targets and their unweighted macro average (lower is better). Right: fixed 95% percentile intervals from 2,000 shared-ID, label-stratified bootstrap resamples of macro-EER differences, conditional on the frozen four-seed ensemble. The figure is display-only: it reads hash-pinned terminal metrics, decision, bootstrap, and provenance but no raw prediction or label table.

Table 3. Per-seed macro EER (%). Each value averages SONAR and ArAD for one frozen seed; the predeclared probability ensemble in Fig. 1 is the primary endpoint.

Method	9101	9102	9103	9104
B1	50.00	50.59	53.78	46.26
B2	53.05	50.67	54.47	47.58
P	53.30	49.26	45.92	41.10

intervals do not resample source voices, training seeds, or a new random-pair draw. We therefore interpret the result as controlled evidence that documented same-item pairing matters beyond the random-pair margin in this protocol, rather than evidence that linguistic content has been causally removed from a representation.

4. PROTOCOL CHECKS AND SCOPE

The comparison has four deliberate safeguards. First, B1, B2, and P share the complete matched-only source pool, so P is not given extra examples or an easier class mix. Second, B2 reuses every P spoof endpoint, the rank-edge count, the selected λ , and the split/language/source/generator marginals; it changes the bona-fide partner from same-group to different-group. Third, the public checkpoint is excluded and each result is from a fresh initialization, avoiding undeclared pretraining. Fourth, the source checkpoint ledger is complete before either target is materialized; target

records omit labels until all raw predictions are emitted. The evaluator also byte-verifies every copied target payload and rejects exact canonical source-target waveform duplicates. These checks do not prove that source and target have no speaker, text, or generator lineage in common, but they rule out the defined direct-content and selection failures.

The result is not a generic claim for pairwise learning. B2 is sometimes worse than BCE (52.01% versus 51.78% macro EER), so the observed gain is not explained by adding Eq. (2) alone. Nor is it a claim that P is the best anti-spoofing model: the absolute EERs remain high. The appropriate interpretation is a controlled interaction in one source/architecture/target configuration: documented natural-synthetic pairing provides useful supervision beyond a random opposite-class partner under the fixed protocol.

For reproducibility, the public lab260ru GitHub organization hosts repository “icassp_antispoofing” at immutable tag “h9pcr-d1.” It contains the source code, frozen protocol, compact ledgers, and result note under the paired- counterfactual experiment directory. The release records source pairing/ exclusion and selection ledgers, checkpoint hashes, separate target record/ label manifests, raw-prediction hash, metrics, seed metrics, bootstrap draws, and a decision record; source and target audio remain under their original dataset licenses. The figure reads only four hash-pinned compact terminal artifacts. These materials make the result auditable but

do not turn the completed targets into a tuning set for a future extension.

5. DISCUSSION AND LIMITATIONS

The B2 result is informative: adding a rank loss alone does not reproduce P, despite its matched edge count and strata. This pattern is consistent with the value of pairing natural and synthetic versions of the documented source item. However, the filename-derived content key is not a transcript-level proof of semantic equivalence, and the experiment does not measure which representation attribute changes. It uses one compact architecture, one source dataset, two fixed targets, and a four-seed ensemble. The weak absolute EER and seed heterogeneity limit deployment claims. A new target or architecture replication must be independently predeclared; it cannot tune the completed protocol.

6. CONCLUSION

Same-item pair ranking lowers the locked two-target transfer EER of a fresh compact anti-spoofing detector relative to both same-pool BCE and an equal-edge random-pair ranking control. The result is supported by a source-only selection chain, target-label firewall, zero exact waveform collisions, and ensemble-conditional paired trial-bootstrap intervals. It motivates further independently frozen replications, not a claim of universal robustness or a causal linguistic-content mechanism.

7. REFERENCES

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